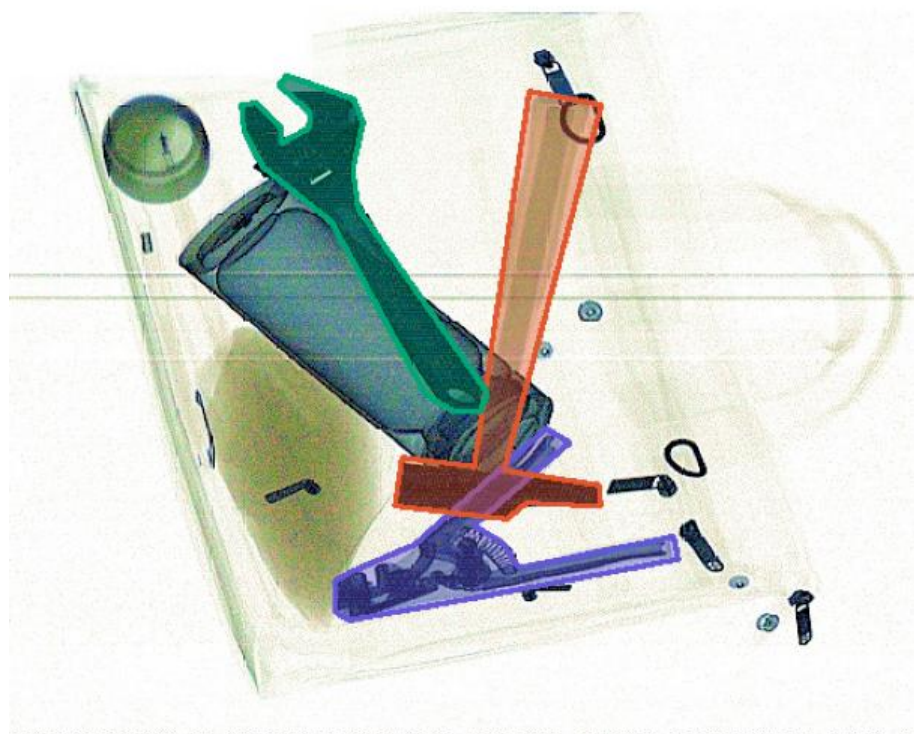


Input



Backbone

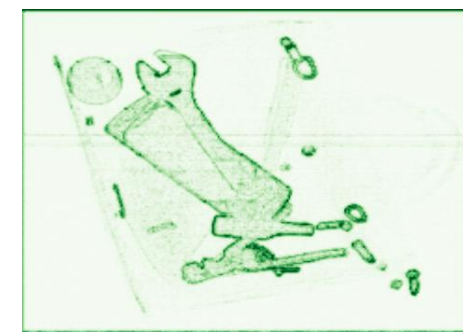
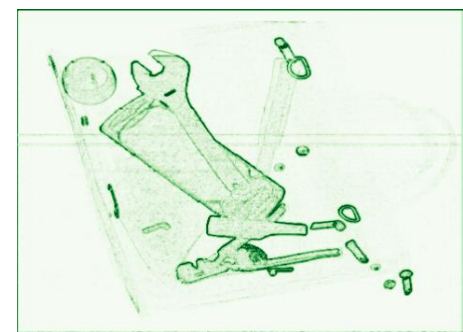
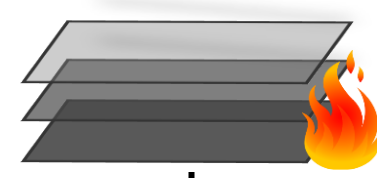
Edge Branch

P0

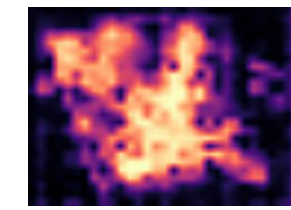
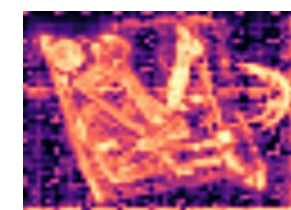
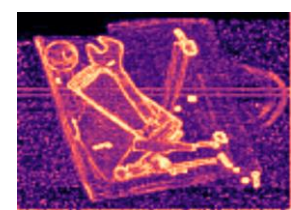
P1

P2

P3



+



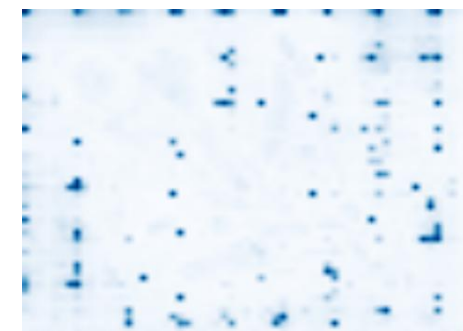
S0

S1

S2

S3

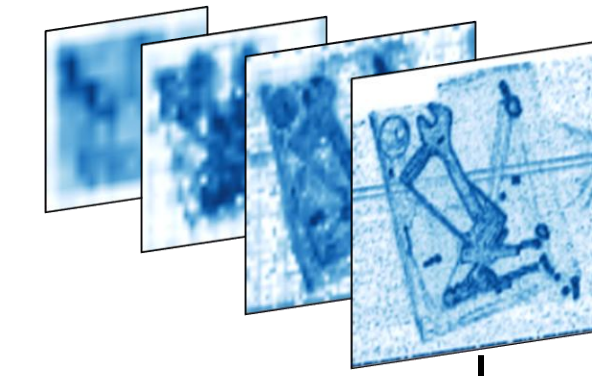
Visual Branch



Detection Head

Channel Mapper

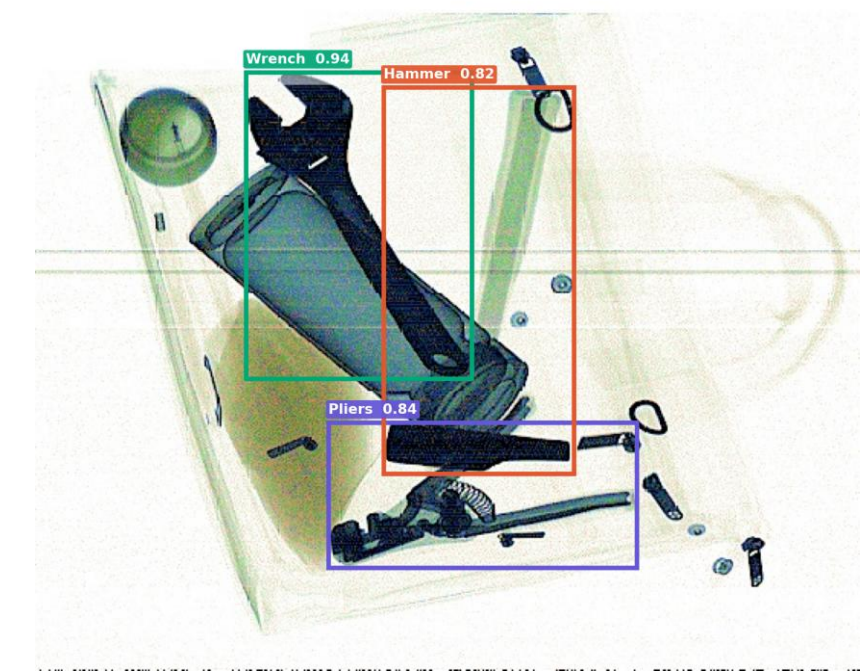
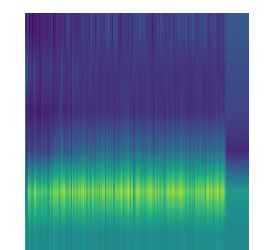
Class Prompts



Gun
Hammer
Knife
Wrench
⋮
Pliers

Multimodal Encoder

Decoder Query Activation

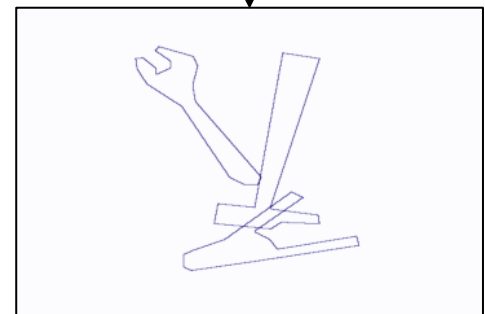


Frozen



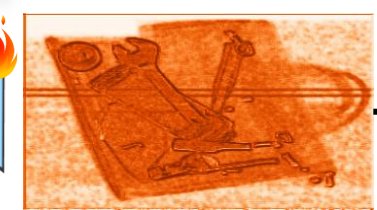
Trained

GT Mask Boundary



$$\mathcal{L}_{\text{boundary}} = -\frac{1}{N} \sum_{i=1}^N \alpha_t (1 - p_{i,t})^{\gamma} \log(p_{i,t})$$

P0

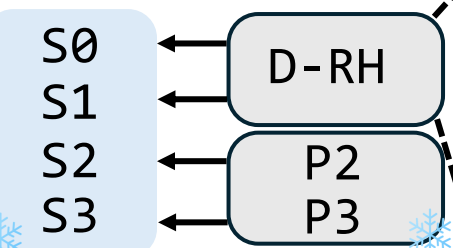


P1



Boundary Pretraining

Residual Finetune



GT labels + Boxes

Detection Head

$$\mathcal{L}_{\text{det}} = \mathcal{L}_{\text{token focal}} + 5\mathcal{L}_{\text{L1 box}} + 2\mathcal{L}_{\text{GIoU}}$$

P0

P1

